

EMIR CAD Engineering Co-op Interview – 9/17/2026

1. Introduce yourself to me. Tell me about your academic coursework, interests, projects, activities, as well as what you enjoy outside of academia.
2. Explain a high-level overview of the ASIC design flow.
3. Draw a cross sectional diagram of an N-channel MOSFET.
 - a. Label each part, including the substrate and wells.
 - b. What should the body be tied to and why?
 - c. What happens if the body is tied the wrong thing and why? What is this effect called?
 - d. What regions can the MOSFET operate in?
 - e. Write down the equations that indicate the conditions for the operation of the device in each of these regions.
 - f. Draw the i_{ds} and V_{ds} curve for different V_g 's applied and explain the difference between the sections of the curve.
4. Draw a schematic for a simple CMOS inverter.
 - a. Explain what happens for each input scenario. Identify the output of the circuit and explain why. (note: make sure to point out why CMOS inverter is necessary rather than a single transistor inverter. This has to do with preventing a floating state condition).
 - b. Draw a VTC (voltage transfer characteristic) curve. Place V_{in} on the x-axis and V_{out} on the y-axis.
 - c. What happens if the PMOS is operating in the triode region and the NMOS is operating in the saturation region? Is this desirable (why or why not)?
5. Redraw the CMOS inverter schematic exactly as before but with the PMOS and NMOS transistor positions flipped.
 - a. What does this circuit do?
 - b. A circuit for this functionality is typically built by cascading two inverters. Why is it desirable to use two cascaded inverters rather than one of this particular circuit?
6. What are the different types of analyses done to find the power characteristics of a transistor?
 - a. What events cause us to evaluate each of these analyses?
 - b. What equations are used to calculate these different power metrics?